MOR – 2025 – (MS)^{2}SC

MOR practical session
Session 1: data decomposition and compression (a-posteriori)

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Instructions

Two practical sessions:

- First session: data decomposition and compression (4h)
- Second session: solving a PDE using PGD (8h)

Practical work will be carried out using Matlab software.

You can work in pairs, but both must code!

A single report will be requested at the end of the two practical sessions, containing a summary of the methods (algorithms, equations, etc.) and results obtained during the two sessions.

SVD approximation of a matrix

600

600

600 → $\underline{\underline{M}}$ (*gray level matrix*)

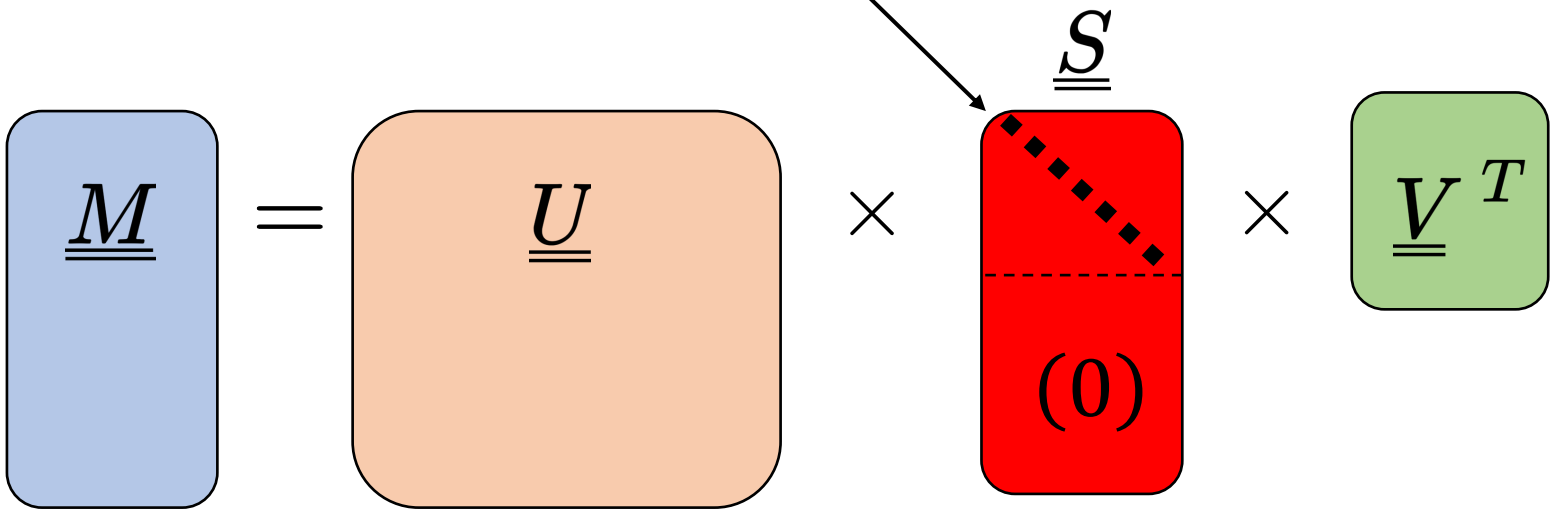
Singular Values Decomposition (SVD)

$$\underline{\underline{M}} = \underline{\underline{U}}\,\underline{\underline{S}}\,\underline{\underline{V}}^T$$

left eigenvectors

right eigenvectors

singular values



$m \times n$ $\quad$ $m \times m$ $\quad$ $m \times n$ $\quad$ $n \times n$

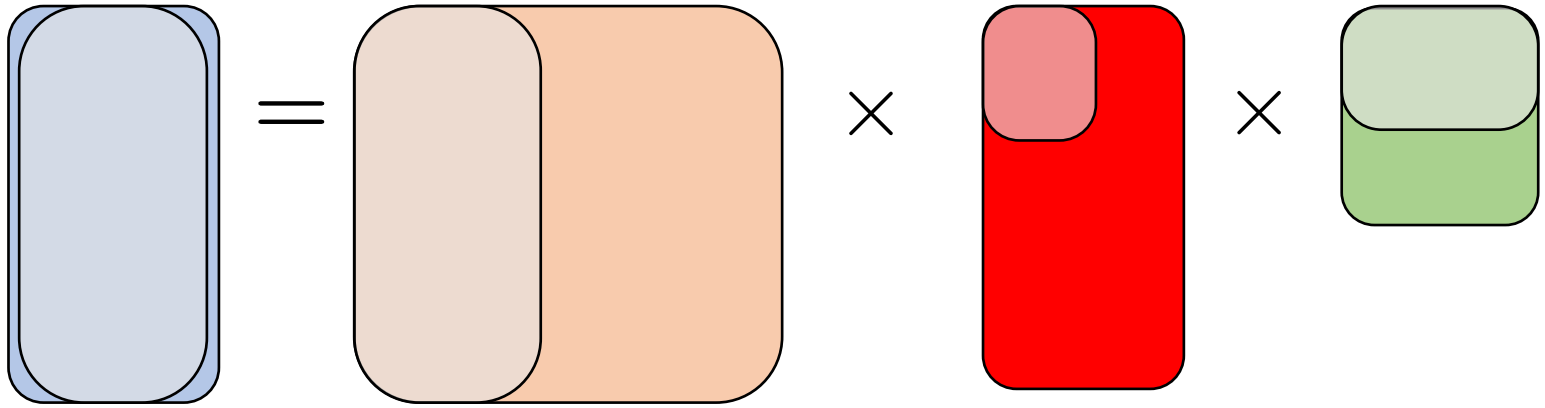
SVD approximation of a matrix

On Matlab:

```
[U,S,V] = svd(M);
```

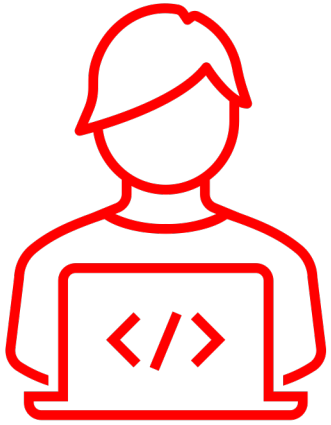
Choose a number of modes: N

With a few modes we can reconstruct the original image: $\underline{\underline{M}}_{\text{POD}} = \sum_{i=1}^{N} S_i \underline{U}_i \underline{V}_i^\top$



$m \times n$ $\quad$ $m \times N$ $\quad$ $N \times N$ $\quad$ $N \times n$

SVD approximation of a matrix



1. Calculate for a number N of POD modes the storage gain achieved by this approach.

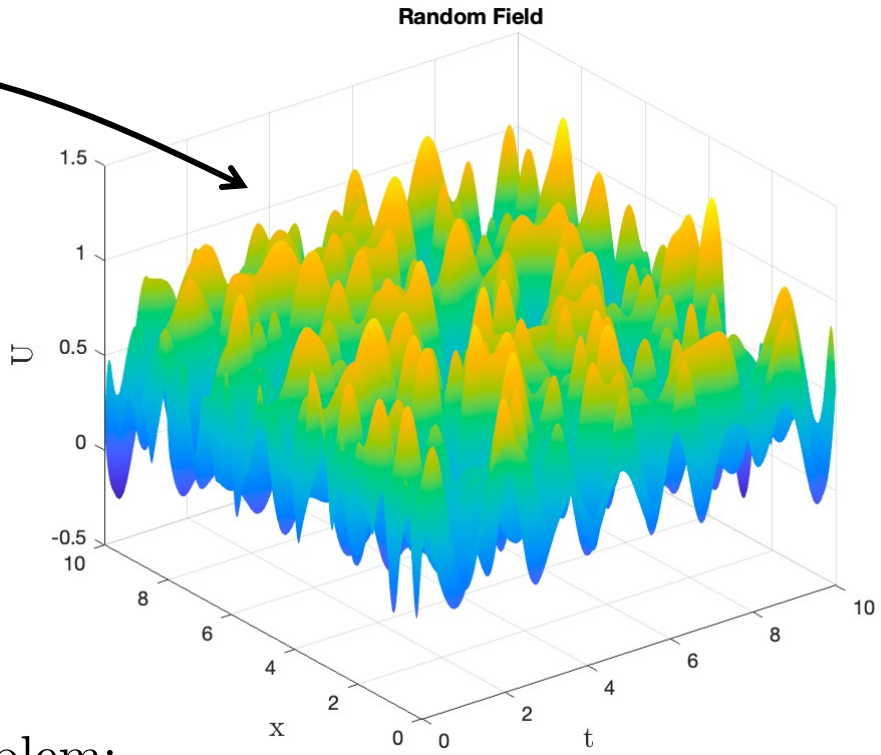
2. Plot the evolution of the relative reconstruction error as a function of the number of modes.

$$\frac{\|\underline{\underline{M}}^{\mathrm{POD}} - \underline{\underline{M}}\|}{\|\underline{\underline{M}}\|}$$

3. Deduce an appropriate number of modes and the effective gain achieved with the previous question.

Random space-time field

$$\begin{array}{rcl} U : \mathcal{W} \times \mathcal{I} & \longrightarrow & \mathbb{R} \\ (x,t) & \mapsto & U(x,t) \end{array}$$



Matrix $N_x \times N_t$

vector $N_x \times 1$

vector $1 \times N_t$

$$\underline{\underline{U}}(x,t) \approx \underline{\underline{U}}_m(x,t) = \sum_{i=1}^{m} \underline{\Lambda}_i(x) \underline{\Gamma}_i^\top(t)$$

number of modes

We are looking for the solution of the following minimization problem:

$$\begin{cases} \underline{\Gamma}_i, \underline{\Lambda}_i = \arg\min_{\mathcal{I}\times\mathcal{W}} \left(\left\| \underline{\underline{U}} - \underline{\underline{U}}_{\text{PGD}}^{i-1} - \underline{\Lambda}_i \underline{\Gamma}_i^\top \right\|^2 \right) \\ \underline{\underline{U}}_{\text{PGD}}^{i-1} = \sum_{k=1}^{i-1} \underline{\Lambda}_k(x) \underline{\Gamma}_k^\top(t) \end{cases}$$

$\longrightarrow$

Problem on $\underline{\Lambda}_i$

Fixed point algorithm

Problem on $\underline{\Gamma}_i$

cf. your lesson !

Course reminder

Intervals:

- $\boldsymbol{X} \in \boldsymbol{\Omega}$, a non-empty bounded domain, where $\boldsymbol{\Omega} \subset \mathbb{R}^d \, (d = 1, 2, 3)$
- $\boldsymbol{I} = [0, T]$ associated with the temporal variable t; we also define: $\| \bullet \|_{\boldsymbol{I}} = \left(\int_{\boldsymbol{I}} \bullet^2 \, \mathrm{d}t \right)^{1/2}$

Vector spaces:

- $\mathcal{W} \equiv \mathcal{H}^1(\boldsymbol{\Omega}) = \left\{ \boldsymbol{u} \in \boldsymbol{L}^2(\boldsymbol{\Omega}), \nabla \boldsymbol{u} \in \boldsymbol{L}^2(\boldsymbol{\Omega}) \right\}$, with $\| \bullet \|_{\mathcal{W}} = \left(\int_{\boldsymbol{\Omega}} \bullet^2 \, \mathrm{d}\boldsymbol{\Omega} \right)^{1/2}$ as the $\boldsymbol{L}^2$ norm of $\mathcal{W}$
- $\mathcal{V} = \mathcal{H}^1_0(\boldsymbol{\Omega}) = \left\{ \boldsymbol{u} \in \mathcal{H}^1(\Omega) \mid \boldsymbol{u}_{|\partial_u \Omega} = \boldsymbol{0} \right\}$
- $\mathcal{U} \equiv \boldsymbol{L}^2(\boldsymbol{I}, \mathcal{W}) = \left\{ \boldsymbol{u} : \boldsymbol{I} \to \mathcal{W}, \int_{\boldsymbol{I}} \|\boldsymbol{u}\|^2_{\mathcal{W}} \, \mathrm{d}t < +\infty \right\}$ where we define: $\| \bullet \|_{\mathcal{U}} = \left(\int_{\boldsymbol{\Omega} \times \boldsymbol{I}} \bullet^2 \, \mathrm{d}\boldsymbol{\Omega} \, \mathrm{d}t \right)^{1/2}$
- $\mathcal{I} \equiv \boldsymbol{L}^2(\boldsymbol{I}, \mathbb{R})$

Course reminder

$$(*)\begin{cases} \boldsymbol{\Lambda}_i, \boldsymbol{\Gamma}_i = \arg\min_{\mathcal{I}\times\mathcal{V}} \left(\left\| \underbrace{\boldsymbol{U}(\boldsymbol{X},t) - \boldsymbol{U}_{i-1}(\boldsymbol{X},t)}_{\text{approximation error of rank } i-1} - \underbrace{\boldsymbol{\Lambda}_i(\boldsymbol{X})\boldsymbol{\Gamma}_i(t)}_{\text{new approximation of rank 1}} \right\|_{\mathcal{U}}^2 \right) \\ \boldsymbol{U}_{i-1}(\boldsymbol{X},t) = \sum_{k=1}^{i-1} \boldsymbol{\Lambda}_k(\boldsymbol{X})\boldsymbol{\Gamma}_k(t) \end{cases}$$

Exploiting the stationarity of the minimization problem in Equation (*) and setting $\boldsymbol{\Delta U} = \boldsymbol{U} - \boldsymbol{U}_{i-1}$, we obtain the coupled problem:

$$\begin{cases} \forall \boldsymbol{\Gamma}^* \in \mathcal{I}, \int_I \boldsymbol{\Gamma}^* \left(\int_\Omega (\boldsymbol{\Delta U} - \boldsymbol{\Lambda}_i\boldsymbol{\Gamma}_i)\boldsymbol{\Lambda}_i \mathrm{d}\Omega \right) \mathrm{dt} = 0 \iff \boldsymbol{\Gamma}_i = \frac{\int_{\mathcal{W}} \boldsymbol{\Delta U}\boldsymbol{\Lambda}_i d\Omega}{\int_{\mathcal{W}} \boldsymbol{\Lambda}_i^2 d\Omega} = f_u(\boldsymbol{\Lambda}_i) \\ \forall \boldsymbol{\Lambda}^* \in \mathcal{V}, \int_\Omega \boldsymbol{\Lambda}^* \left(\int_I (\boldsymbol{\Delta U} - \boldsymbol{\Lambda}_i\boldsymbol{\Gamma}_i)\boldsymbol{\Gamma}_i \mathrm{dt} \right) \mathrm{d}\Omega = 0 \iff \boldsymbol{\Lambda}_i = \frac{\int_I \boldsymbol{\Delta U}\boldsymbol{\Gamma}_i \mathrm{dt}}{\int_I \boldsymbol{\Gamma}_i^2 \mathrm{dt}} = g_u(\boldsymbol{\Gamma}_i) \end{cases}$$

The functions are defined and unique up to a multiplicative factor. To ensure the uniqueness of the solution in (*) , the spatial function is usually normalized.

First step: calculating integrals

How to compute $\displaystyle\int_0^L a(x)\cdot b(x)dx$?

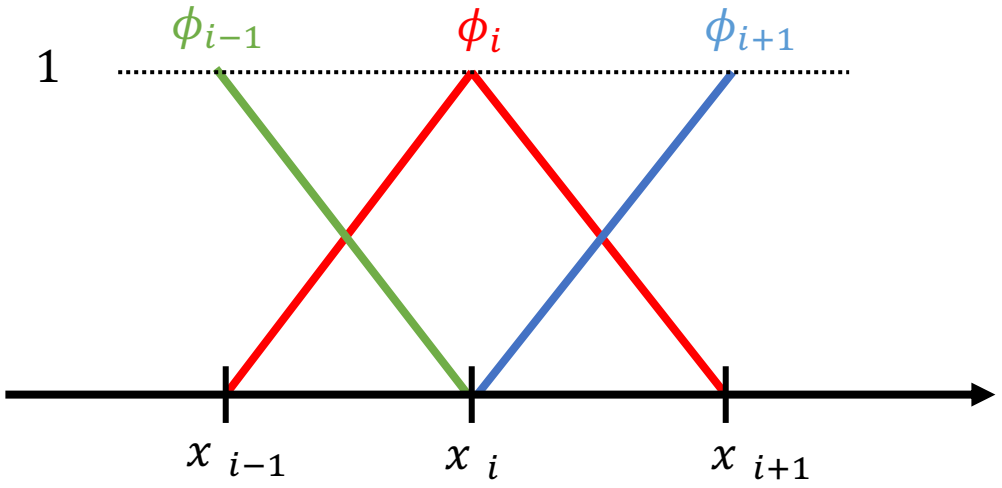
(We apply the same approach for the *temporal integrals*.)

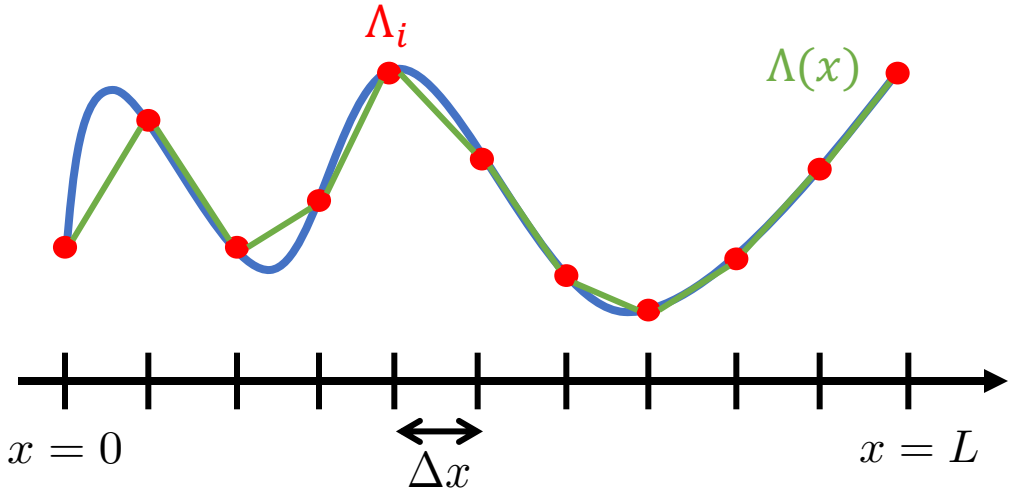
The $\underline{\Lambda}$ and $\underline{\Gamma}$ are discrete fields defined at points.

To compute the integrals we will interpolate these discrete fields using **shape functions** (in the same way as for the finite element method). For this tutorial we will use the simplest shape functions : **piecewise linear**.

In this way, we have :

$$\underline{\Lambda}(x) = \sum_{i=1}^{N_x} \phi_i(x) \times \Lambda_i \text{ with } \begin{cases} \Lambda_i = \Lambda(x_i) \\ x_i = i \times \Delta x \end{cases}$$





In our case the shape functions are written as :

$$\begin{cases} \phi_i(x_i) = 1 \\ \phi_i(x) = \frac{x - x_{i-1}}{\Delta x} \text{ if } x \in [x_{i-1}; x_i] \\ \phi_i(x) = \frac{x_{i+1} - x}{\Delta x} \text{ if } x \in [x_i; x_{i+1}] \\ 0 \quad \text{otherwise} \end{cases}$$

Thus, with the use of these shape functions, the integral of a product of functions is written as :

$$\int_0^L a(x) \cdot b(x) dx = \sum_{i=0}^{N_x-1} \int_{x_i}^{x_{i+1}} a(x) \cdot b(x) dx$$

With :

$$a(x) = \sum_{j=1}^{N_x} \phi_j(x) a_j$$

$$b(x) = \sum_{j=1}^{N_x} \phi_j(x) b_j$$

Taking advantage of the fact that only functions of form ϕ_i and ϕ_{i+1} are nonzero on the interval $[x_i; x_{i+1}]$ (they are identically zero everywhere outside) the previously defined integral becomes :

$$\sum_{i=0}^{N_x-1} \int_{x_i}^{x_{i+1}} \left(a_i\phi_i + a_{i+1}\phi_{i+1}\right) \times \left(b_i\phi_i + b_{i+1}\phi_{i+1}\right) dx$$

in algebraic form :

$$\begin{pmatrix} a_i \\ a_{i+1} \end{pmatrix}^T \cdot \begin{pmatrix} \phi_i^2 & \phi_i\phi_{i+1} \\ \phi_i\phi_{i+1} & \phi_{i+1}^2 \end{pmatrix} \cdot \begin{pmatrix} b_i \\ b_{i+1} \end{pmatrix}$$

constant

$$\int_{x_i}^{x_{i+1}} \bullet \, dx = \begin{pmatrix} a_i \\ a_{i+1} \end{pmatrix}^T \cdot \underbrace{\begin{pmatrix} \int_{x_i}^{x_{i+1}} \phi_i^2 dx & \int_{x_i}^{x_{i+1}} \phi_i \phi_{i+1} dx \\ \int_{x_i}^{x_{i+1}} \phi_i \phi_{i+1} dx & \int_{x_i}^{x_{i+1}} \phi_{i+1}^2 dx \end{pmatrix}}_{\underline{\underline{m}}} \cdot \begin{pmatrix} b_i \\ b_{i+1} \end{pmatrix}$$

constant

(mass) elementary matrix

Let us note :

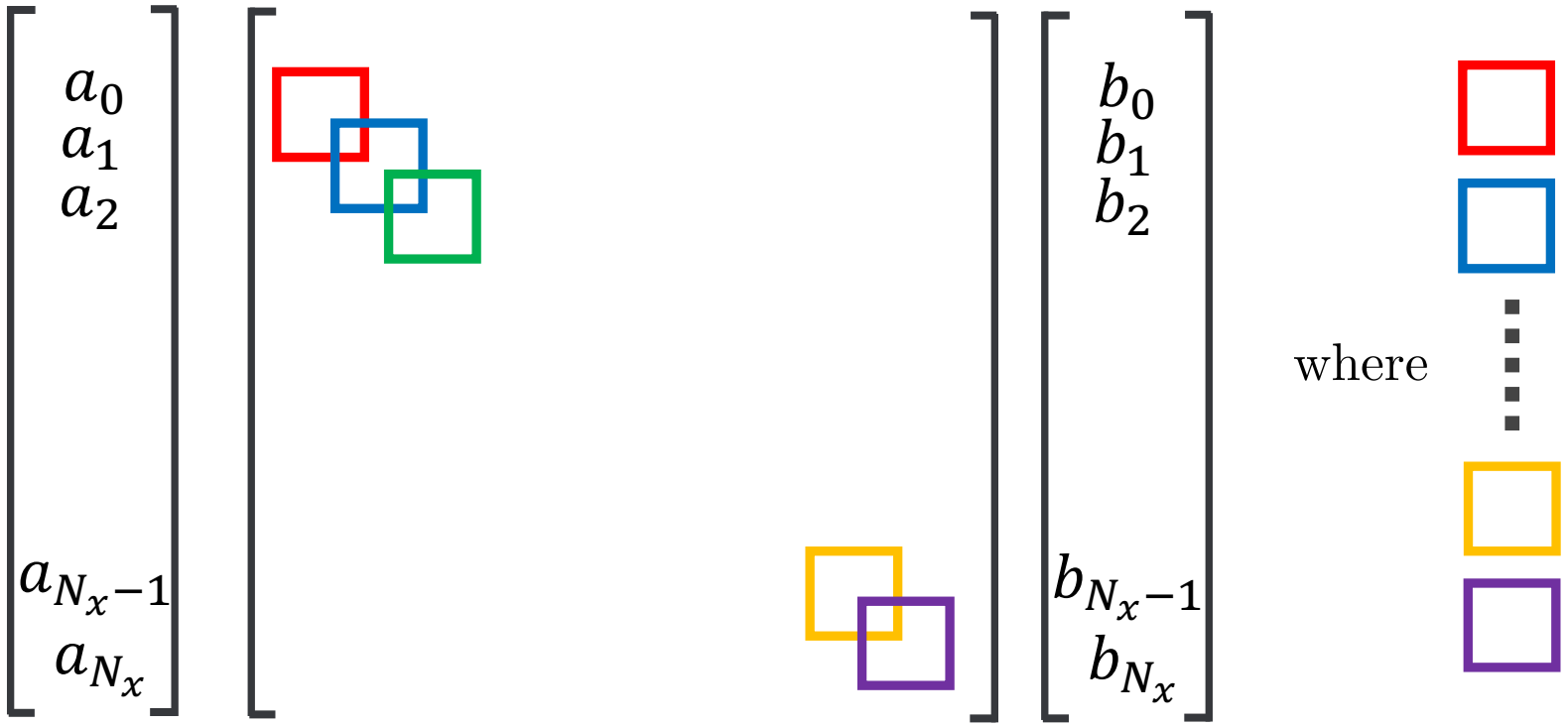
$$\int_{x_i}^{x_{i+1}} \phi_i^2 dx = \int_{x_i}^{x_{i+1}} \phi_{i+1}^2 dx$$

Show that :

$$\underline{\underline{m}} = \Delta x \cdot \begin{pmatrix} 1/3 & 1/6 \\ 1/6 & 1/3 \end{pmatrix}$$

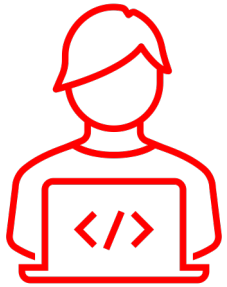
Calculation of integrals by the finite element method

(Here, we have $N_x + 1$ nodes and N_x elements...)



where $= \frac{\Delta x}{6}\begin{pmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{1} & \mathbf{2} \end{pmatrix}$

$$\int_0^L a(x)b(x)\mathrm{dx} = \underline{a}^\top \underline{\underline{I}}_x \underline{b} \qquad \int_0^T c(t)d(t)\mathrm{dt} = \underline{c}^\top \underline{\underline{I}}_t \underline{d}$$



<u>In this case we have two strategies:</u>

- either we assemble the elementary matrices (cf. FEM)
- or we build directly the operator in one shot (faster and easier by taking advantage of the matlab functions : *diag* & *ones*)

Second step: the fixed-point algorithm

1. Calculating the field to be approximated

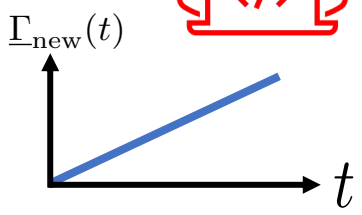
$$\Delta \underline{\underline{U}} = \underline{\underline{U}}_f - \underline{\underline{U}}_{i-1}$$

2. Initialize fixed-point stagnation criterion

$$s = 100$$

3. **Initialize** the "new" time function

$$\underline{\Gamma}_{\text{new}} = \ldots$$



4. As long as the stagnation criterion is not satisfied:

$$s \leqslant 10^{-3}$$

   1. Store time function in "old" variable

$$\underline{\Gamma}_{\text{old}} = \underline{\Gamma}_{\text{new}}$$

   2. Compute spatial function + **normalize**

$$\underline{\Lambda}_{\text{new}} = \ldots \quad N_x \times \text{ integration on time}$$

   3. Compute new time function "new"

$$\underline{\Gamma}_{\text{new}} = \ldots \quad N_t \times \text{ integration on space}$$

   4. Calculate stagnation criterion between "old" and "new" time functions

$$s = \frac{\int_I \left(\underline{\Gamma}_{\text{new}} - \underline{\Gamma}_{\text{old}}\right)^2 dt}{\int_I \left(\underline{\Gamma}_{\text{old}}\right)^2 dt}$$

1. Store new modes

$$\underline{\Lambda}_{\text{i}} = \underline{\Lambda}_{\text{new}}$$

$$\underline{\Gamma}_{\text{i}} = \underline{\Gamma}_{\text{new}}$$

Third step: error calculation

$$\text{Error} = \frac{\left\| \underline{\underline{U}} - \underline{\underline{U}}^{\mathrm{PGD}} \right\|_{\mathcal{U}}}{\left\| \underline{\underline{U}} \right\|_{\mathcal{U}}} \longrightarrow \text{Error} = \frac{\sqrt{\int_{\mathcal{W}} \int_{\mathcal{I}} \left(\underline{\underline{U}} - \underline{\underline{U}}^{\mathrm{PGD}} \right)^2 \, \mathrm{d}t \, \mathrm{d}\Omega}}{\sqrt{\int_{\mathcal{W}} \int_{\mathcal{I}} \left(\underline{\underline{U}} \right)^2 \, \mathrm{d}t \, \mathrm{d}\Omega}}$$

How to calculate this error numerically?

1. First we integrate on $\mathcal{I}$, we obtain a vector of size N_x
2. Then we integrate on $\mathcal{W}$, we get a **scalar**.

Plot the evolution of the relative reconstruction error as a function of the number of modes.

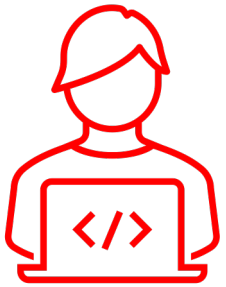
Final step: orthogonalize spatial modes

Why orthogonalize the modes?

- To avoid redundancy of information between modes
- This will be useful for TP 2 for the « Update phase »

How to orthogonalize the modes?

- Gram-Schmidt algorithm!



We want: $\langle \underline{\Lambda}_i, \underline{\Lambda}_j \rangle = \delta_{ij}$ $\quad \langle \circ, \square \rangle = \int_{\mathcal{W}} \circ \square \, \mathrm{d}\boldsymbol{\Omega}$

Consider that we have, after the fixed point algorithm: $\underline{\Lambda}_{m+1}^{\text{new}}$

We apply the Gram-Schmidt procedure:

$$\underline{\Lambda}_{m+1} = \underline{\Lambda}_{m+1}^{\text{new}} - \sum_{i=1}^{m} \langle \Lambda_{m+1}^{\text{new}}, \underline{\Lambda}_i \rangle \cdot \underline{\Lambda}_i$$

In this way we have:

$$\forall i \in [1, m], \langle \underline{\Lambda}_{m+1}, \underline{\Lambda}_i \rangle = 0$$

However, as we « remove » information at the level of $\underline{\Lambda}_{m+1}^{\text{new}}$, we have to « add » it in $\underline{\Gamma}_i$:

$$\underline{\Gamma}_i \leftarrow \underline{\Gamma}_i + \underline{\Gamma}_{m+1} \cdot \langle \underline{\Lambda}_{m+1}^{new}, \underline{\Lambda}_i \rangle, \forall i \in [1, m]$$

Prove it for m=1

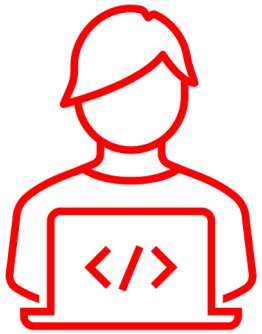
Finally we normalize:

$$\underline{\Lambda}_{m+1} \leftarrow \frac{\underline{\Lambda}_{m+1}}{\sqrt{\int_{\mathcal{W}} \left(\underline{\Lambda}_{m+1}\right)^2 \, \mathrm{d}\boldsymbol{\Omega}}}$$

$$\underline{\Gamma}_{m+1}^{\text{new}} \leftarrow \underline{\Gamma}_{m+1}^{\text{new}} \times \sqrt{\int_{\mathcal{W}} \left(\underline{\Lambda}_{m+1}\right)^2 \, \mathrm{d}\boldsymbol{\Omega}}$$

Back to the image

Implement the PGD algorithm for the approximation of the Maria Callas image.



Compare the SVD decomposition and the PGD decomposition.
What are the advantages/disadvantages of these two methods?

Calculate the algorithmic complexity associated with these methods.

<u>To go further:</u> apply these different methods to color images (RGB) of your choice